

Predicting Movie Titles

by Kyle Rysdam

Outline

- Task
- Data
- Baseline
- Models
 - Architecture
 - Dev set performance
 - Test set performance
- Future directions

The task: Title prediction

“Thomas A. Anderson is a man living two lives. By day he is an average computer programmer and by night a hacker known as Neo. Neo has always questioned his reality, but the truth is far beyond his imagination. Neo finds himself targeted by the police when he is contacted by Morpheus, a legendary computer hacker branded a terrorist by the government. As a rebel against the machines, Neo must confront the agents: super-powerful computer programs devoted to stopping Neo and the entire human rebellion.”

→ **“The Matrix”** 

→ **“Neo” “The One” “A Rebel Against the Machines”** 

The data

- 54,000 movies
- Summaries (Average length: 101 tokens)
- Titles (Average length: 2.3 tokens)



RADMIRKAZ · UPDATED 2 YEARS AGO

Genre Classification Dataset IMDb

Test your NLP text classification skills

Baseline

- Model score = avg. cos. similarity of prediction to real next token
 - Using word embeddings
 - On a scale from -1.0 to +1.0
- Baseline model: always predict average embedding of all training titles
 - Close to ["The" <s> FILL FILL FILL FILL FILL FILL FILL FILL]
- Baseline score (50-dimensional embeddings) = **0.68**
- Baseline score (300-dimensional embeddings) = **0.65**

The models

- Encoder-decoder RNNs + LSTM
- **Summary** → Embeddings → Encoder → Decoder → Embeddings → **Title**

The models

Name:	<i>Simple</i>	<i>Fancy</i>
Summary embeddings:	word	sentence
Embeddings size:	50 dim	300 dim
Hidden layer units:	32 units	256 units
Parameters:	~24,000	~1,200,000
Time to train:	30 minutes	10 minutes

Dev set scores

Name:	<i>Simple</i>	<i>Fancy</i>
Baseline similarity:	0.68	0.65
Model similarity:	0.7988	0.7613
Compared to baseline:	+17%	+17%
Accuracy in dev set:	0 / 10,000	0 / 10,000

Test set predictions

Real title	Simple's prediction	Fancy's prediction
Snow White and the Seven Dwarfs		
The Matrix		
Star Wars: Episode IV – A New Hope		
Memento		
Portrait of a Lady on Fire		
Inception		
Coco		
Shrek		
Finding Nemo		
Psycho		

Good title for this movie

Coherent title

Nonsense

Test set predictions

Real title	Simple's prediction	Fancy's prediction
Snow White and the Seven Dwarfs	Little Little Little	The Love of the Beautiful
The Matrix	One One One	The One
Star Wars: Episode IV – A New Hope	The Beast One	The Man One
Memento	One Life. One	The One
Portrait of a Lady on Fire	La Dans La La	La La La La
Inception	One One	The One
Coco	La Voz	La De La Noche
Shrek	Dragon Cat	The Man But So So
Finding Nemo	The Ghost the Ghost	The One One
Psycho	One One One	The Just Just

Good title for this movie

Coherent title

Nonsense

Test set predictions

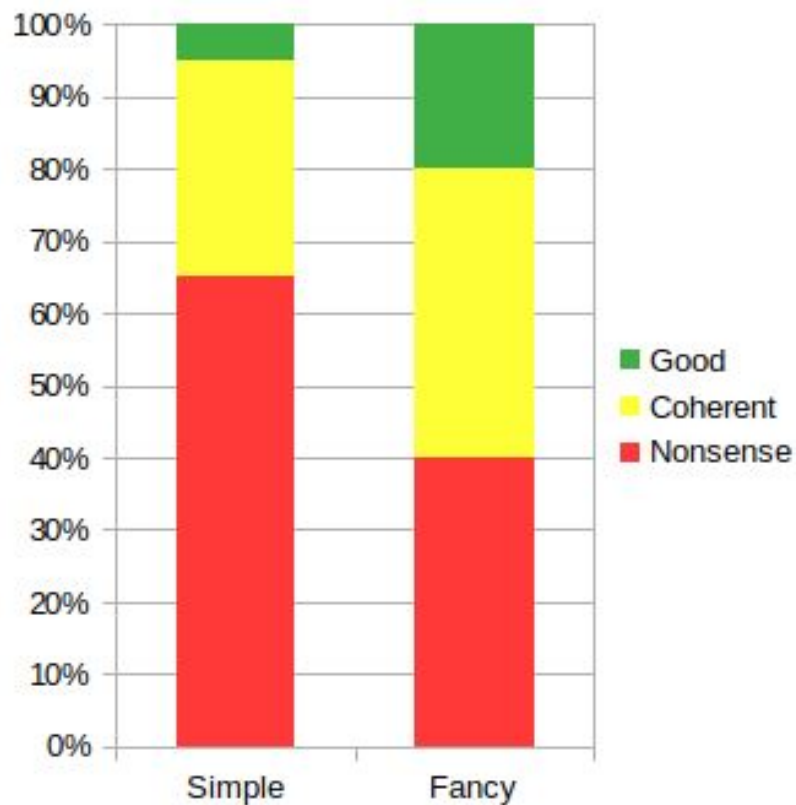
Real title	Simple's prediction	Fancy's prediction
The Godfather	One One One	The Man
Cloud Atlas	The One One	The One
The Wizard of Oz	The Ghost One	The Man One
The Lord of the Rings: The Fellowship of the Ring	The Ghost the Ghost	The Ghost of the Ghost
Citizen Kane	One One One	The One
Arrival	The One One	The One
Spirited Away	Love One Life	The So So So
Gone with the Wind	One Life. One	The One One
WALL-E	Into the Into	The Just Just Just
The Danish Girl	"One One One	Der Und Und Und Und Und

Good title for this movie

Coherent title

Nonsense

Test set predictions



Future directions

- Use more data (50,000 movies isn't enough to learn English)
- Fine-tune a pre-trained language model like GPT-2
- If those work well, try the reverse task: title → summary

Questions?

Thank you!